

Research Summary

Artificial Life Simulation for Emergent Adaptive Behavior

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Research question

Can adaptive, knowledge-like behavior emerge from an agent's interaction with a simulated ecology, when the agent is not given predefined semantic labels, explicit instructions about resource use, or a hand-authored task-reward function?

Honest one-line status: the project has solid, controlled evidence for experience-based learning and ecological interaction, and an honestly unsolved problem in long-run population sustainability. It does not yet claim intentional farming, social transmission, or open-ended evolution.

Part 1 — What the project has established (strengths, with reason + data)

Each item gives the claim, why it is credible (the control or method), and the data, plus its scope limit. The project rule: a pattern is not promoted to ‘knowledge’ until controls rule out simpler explanations (food hotspots, hunger pressure, deterministic movement, shared corridors).

1. A working ecology substrate exists — not just an idea.

A full seed → plant → fruit → consumed cycle runs and is logged in a tuned 100x100 world. Data: Phase 1 passed 5/5 seeds; per run ~139.0 germinated seeds, 33.6 mature plants, 68.6 fruiting events, 47.8 plant-food consumptions. Scope: establishes the substrate only — no cognition claim.

2. Reward-place learning is measurable and large.

A strict leave-then-return metric is compared against a random-position baseline, so ‘returning to food’ cannot be explained by chance position. Data: Phase 2 passed 3/3 seeds; mean owner return lift 41.7x (per-seed 52.7x / 38.7x / 33.7x). Scope: place memory, not semantic understanding.

3. Experience-based food-value learning — the strongest result, hardened with controls + statistics.

The baseline architecture is value-blind at eating time (it eats whatever fits the mouth). When experience-based value memory is enabled, agents learn each food's energy value and stop eating the low-value one. The result survives three independent controls:

- Per-agent causality (A1): of 12 agents, 10 learned to skip the low-value seed — and 0/12 skipped it without first tasting the better food. Behavior change is bound to direct individual experience.
- Mechanism control (A5): same seed, same config, only the memory switch differs. Memory on → 10/12 skip (108 seed meals, 2,163 skips); memory off → 0/12 skip (382 seed meals, 0 skips); plant meals identical 52 = 52. The behavior disappears when the mechanism is removed.
- Quantitative threshold (A2): agents learn seed = 50 << plant = 250, and skip seed because it falls below a learned threshold (125), not because of a ‘bad’ label.

Window	Value-blind (no learning)	Value-learning enabled
0-1k ticks	591	236
1-2k ticks	613	105
2-3k ticks	370	15
3-4k ticks	336	5
4-5k ticks	392	5
5-6k ticks	334	5

Low-value seed meals per 1,000 ticks. Learning is population-wide (10/10 who tasted both foods learned; mean time-to-learn 14.1 ticks, range 2-59), not one lucky agent. Scope: individual learning only — not social transmission.

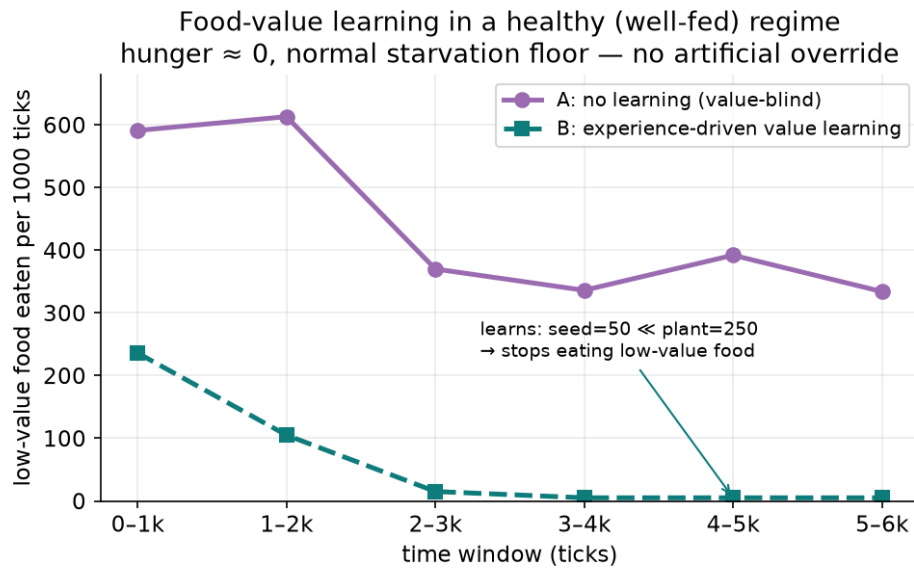


Figure 1. Food-value learning curve: with value-learning, low-value food consumption collapses toward zero while the value-blind baseline stays high.

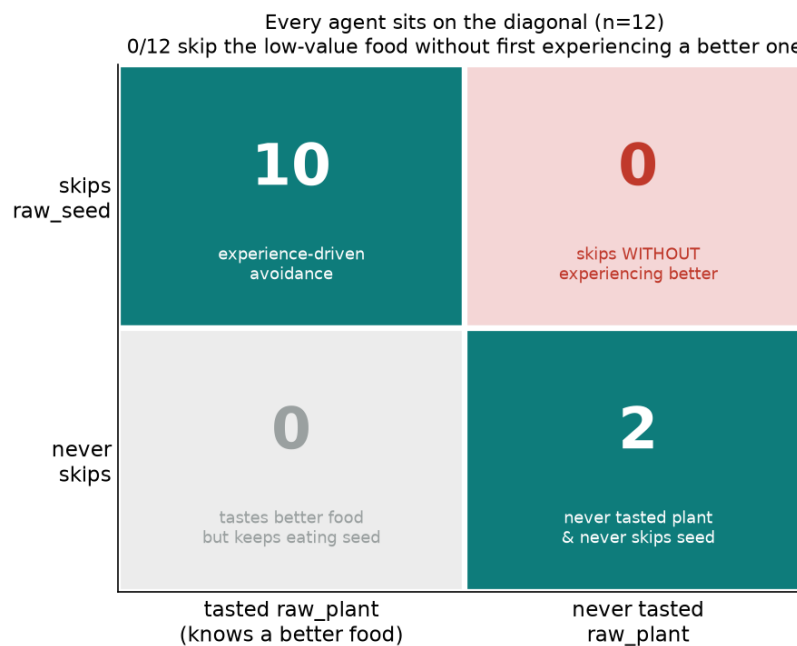


Figure 2. Per-agent causality: every agent that skips the low-value seed has first tasted the better food. The cell 'skips without tasting better food' is empty (0/12).

4. Agent actions can enter delayed ecological chains (proto-farming substrate).

Agent-moved seeds were tracked through the full seed → plant → fruit → consumed chain against a control. Data: Phase 3 passed 3/3 seeds; ~162.3 moved seeds/run, 16.3 completed moved-seed chains, moved/control lift 1.40x. Scope: a proto-farming substrate — explicitly not intentional or understood farming.

5. An evolutionary (neuroevolution) controller works as a proof of concept.

A neural controller (1,281-parameter genome) is evolved by a genetic algorithm with tournament selection and elitism, evaluated across held-out seeds. Data: over 12 generations (population 30), best fitness rose to 30.0. Scope: proof that the neuroevolution loop functions; not yet a large-scale evolved-behavior result.

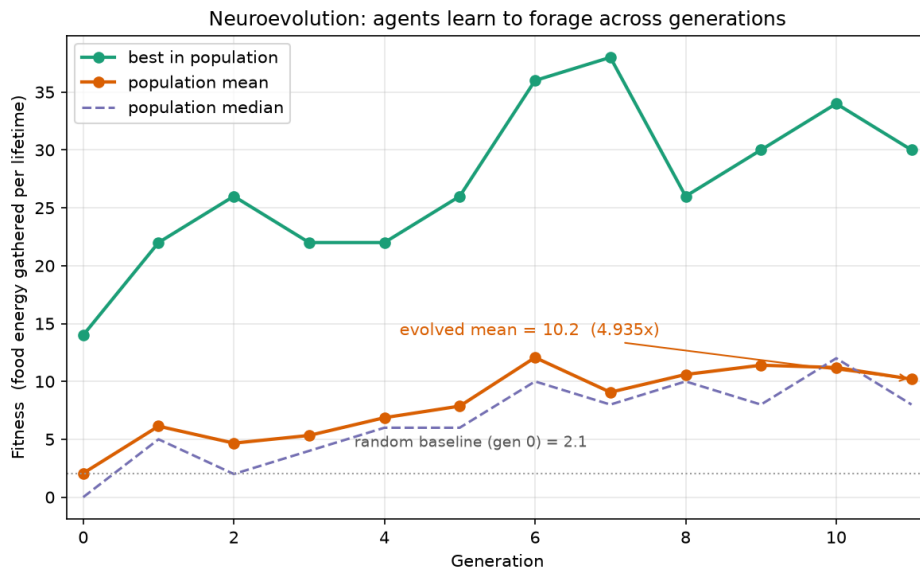


Figure 3. Neuroevolution proof of concept: best/mean fitness improve over generations.

6. Method discipline is itself a result.

67/67 automated tests pass; movement RNG is now bound to the experiment seed (fixed 26 June), making multi-seed replication valid; new instrumentation is opt-in and byte-identical by default. The project treats negative results as evidence and writes an explicit 'can claim / cannot claim' box for every result.

Part 2 — Where the project is honestly stuck (the open problem)

The blocker: a mortal population still goes extinct over the long run. What makes this interesting is that the root cause has been progressively localized with evidence, not hand-waved:

- 1 First suspected fecundity (too few births) → ruled out: female fecundity reached near-replacement.
- 2 Then demographic dynamics → delayed density dependence produced boom-crash oscillation, but tuning it did not save the population.
- 3 Then foraging access → measured directly: agents sense food only within ~4 cells while realistic food is sparser, so at realistic density 0% of agents sense food and intake collapses (drain:intake ~453:1).
- 4 Option GA (28 June) solved foraging access — a seed-rain plant economy woke the plant lifecycle (0 → ~6,800 fruits) and a decoupled ‘smell’ sensing radius raised the sensing fraction 0.04 → 0.77 (meal rate x6.5) on plant-only food, with no artificial crutch. Dispersed agents reached an energy surplus of 200-550. Access is no longer the bottleneck — proven by measured surplus.
- 5 The bottleneck relocated to a deeper layer: spatial-demographic. New telemetry shows the honest failure mode: after a crash, global food looks fine (food-per-capita 5.2 → 19.83) but local food-per-capita around the cluster is 0.0 — agents starve locally while food sits elsewhere.

The current tension: clustering (needed to find mates) → local food depletion → local starvation; dispersal (needed to eat) → mates too far apart → Allee effect, births fade; plus synchronized lifespan death producing a death wave (death-window CV 0.93). Representative baseline: extinct at tick 549, peak population 148, 114 births, deaths = 123 lifespan / 37 starvation / 4 durability.

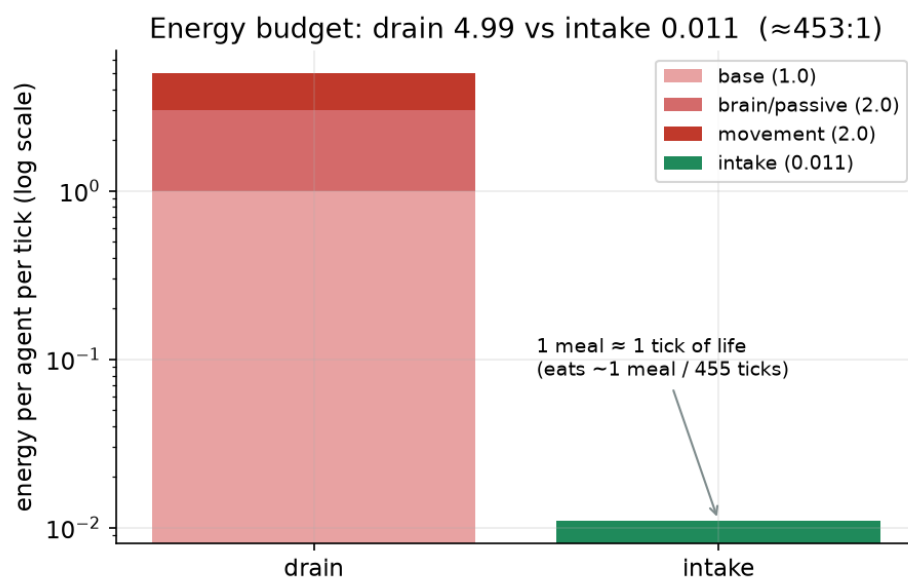


Figure 4. The earlier energy bottleneck (drain vs intake) that the foraging-access fix removed; the residual blocker is now demographic, not energetic.

This is reported as an honest negative. Each ‘failure’ sharpened the question and ruled out an alternative, which is exactly how the real bottleneck was found. Resolving the clustering-vs-dispersal tension and de-synchronizing death on a world with a real carrying capacity is the next experiment, not a solved result.

Part 3 — What the project does NOT claim (to be explicit)

- Intentional farming or semantic understanding of seeds.
- Social transmission of food knowledge (only individual learning is shown).
- Stable open-ended evolution across generations (population not yet sustained).
- Some positive results were isolated in tuned or energy-surplus regimes to separate one mechanism at a time; that is stated wherever it applies.

Part 4 — Where I would value an advisor's guidance

- 1 Population dynamics / theoretical ecology: is the clustering-vs-dispersal tension best framed as a metapopulation / Allee problem, and what minimal mechanism would let a mortal population converge around a real carrying capacity?
- 2 Experiment design & statistics: sharpening the multi-seed confirmatory protocol (effect sizes, confidence intervals, non-parametric tests) so the food-value-learning result is publication-grade.
- 3 Framing & scope: which single result is strongest to lead with for YSC / ISEF and the Junior Young Rising Stars award, and how to scope the claims so they stay defensible.

I have a CV, this summary, and a fully open, reproducible codebase with a complete report archive. I would be grateful for even brief feedback, and would welcome the chance to discuss the project further.

Methods note

Python agent-based artificial-life simulation. The world has a grid environment, food spawning, a plant lifecycle, seeds, energy drain, a metabolism model (food composition → embodied energy, not symbolic reward), day-night and seasonal effects, agent memory, reproduction gates, lineage tracking, and experimental telemetry. Controls include random-position and current-position baselines, memory ablations, state decoupling, mechanism on/off controls, and multi-seed replication. Reproduction is gated by energy and body durability (durability 10 → 0 births; durability 26 → 50 births under the same energy regime), which is why energy surplus alone did not immediately yield births.

Source index (key reports in the repository)

Topic	File
Phase 1-3 results	reports/phase3/phase1_to_phase3_research_success_report_2026-06-13.th.md
Food-value learning	reports/food_value_learning_paper_2026-06-19.th.md
Per-agent food learning	reports/agent_food_value_individual_tracking_2026-06-23.th.md
Statistical hardening (Tier A)	reports/tier_a_completion_summary_2026-06-30.th.md
Energy economy	reports/energy_economy_diagnosis_2026-06-19.th.md
Foraging-access fix (Option GA)	reports/option_ga_foraging_access_results_2026-06-28.th.md
Spatial-demographic blocker (Option KHO)	reports/option_kho_demographic_telemetry_baseline_report_2026-06-28.th.md
Full competition report	reports/competition_full_report_artificial_evolution_2026-06-29.th.md